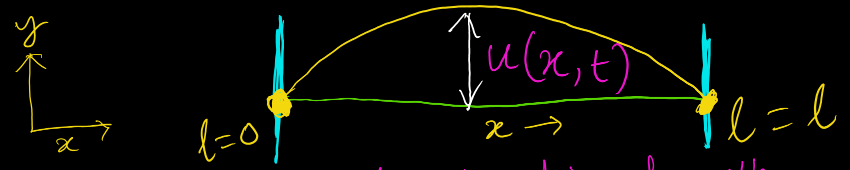
Classical Wave Equation



1D vibrating String.

$u(x,t)$ = Displacement of the string from it's equilibrium horizontal position

= $\boxed{\text{Amplitude}}$

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 u}{\partial t^2} \longrightarrow \text{Classical wave eq}^n\text{:}$$

2nd order Partial differtial eq^n

v → Speed at which a disturbance moves

$x, t \Rightarrow$ Independent
$u(x,t) \Rightarrow$ depenent

Boundary Condition :

$u(0,t)=0$ & $u(l,t)=0$ for all t

Key Step, Assume $u(x,t) = \underbrace{X(x)}_{\text{Space part}}\,\underbrace{T(t)}_{\text{Time part}}$ — Seperation of Variables

Replacing wave eq^n –

$$T(t)\frac{d^2(x)}{dx^2} = \frac{1}{v^2}X(x)\frac{d^2(T)}{dt^2}$$

Deviding $u(x,t) = X(x)\,T(t)$

$$\underbrace{\frac{1}{X(x)}\frac{d^2(x)}{dx^2}}_{\text{function of } x} = \underbrace{\frac{1}{v^2 T(t)}\frac{d^2T}{dt^2}}_{\text{function of } t} = k = -\beta^2$$

$$\frac{d^2X(x)}{dx^2} = -\beta^2 X(x)$$

$$\frac{d^2T(t)}{dt^2} = -\beta^2 v^2 T(t)$$

Solution to this eq^n is ⇒

$$\boxed{\begin{aligned} \text{Space}\quad X(x) &= A\cos(\beta x) + B\sin(\beta x) \\ \text{Time}\quad T(t) &= C\cos(\beta v t) + D\sin(\beta v t) \end{aligned}}$$

General Solution

Vibrating String

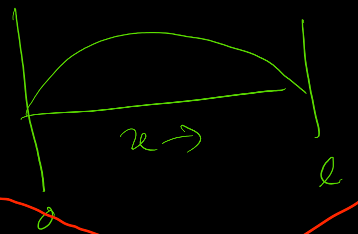
General Solution to the wave equation is a superposition of normal modes

At Boundary condition :-

$$X(0) = 0 = A\cos(\beta \cdot 0) + B\sin(\beta \cdot 0)$$

$$X(\ell) = 0 = B \sin(\beta \ell)$$

$$X(x) = B \sin\left(\frac{n\pi x}{\ell}\right)$$



$$\beta \ell = \sin^{-1}(0) = n\pi$$

$$\beta = \frac{n\pi}{\ell} \quad n \in \mathbb{Z}$$

Quantization

$$T(t) = C\cos\left(\frac{n\pi v t}{\ell}\right) + D\sin\left(\frac{n\pi v t}{\ell}\right)$$

$$= E\cos\left(\frac{n\pi v t}{\ell} + \phi\right) \quad \phi \rightarrow \text{Phase factor}$$

$$u(x,t) = A\cos\left(\frac{n\pi v t}{\ell} + \phi\right)\sin\left(\frac{n\pi x}{\ell}\right)$$

$$\boxed{u(x,t) = \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi v t}{\ell} + \phi_n\right)\sin\left(\frac{n\pi x}{\ell}\right)}$$

$$u(x,t) = \sum_{n=1}^{\infty} A_n u_n(x,t)$$

$\{u_n(x,t)\} \rightarrow$ "normal modes"

"Standing waves"

Each $\{u_n(x,t)\} \rightarrow$ "normal modes"
"Standing waves"

Because x, t is not changing, node position remains same

Spatial dependence:

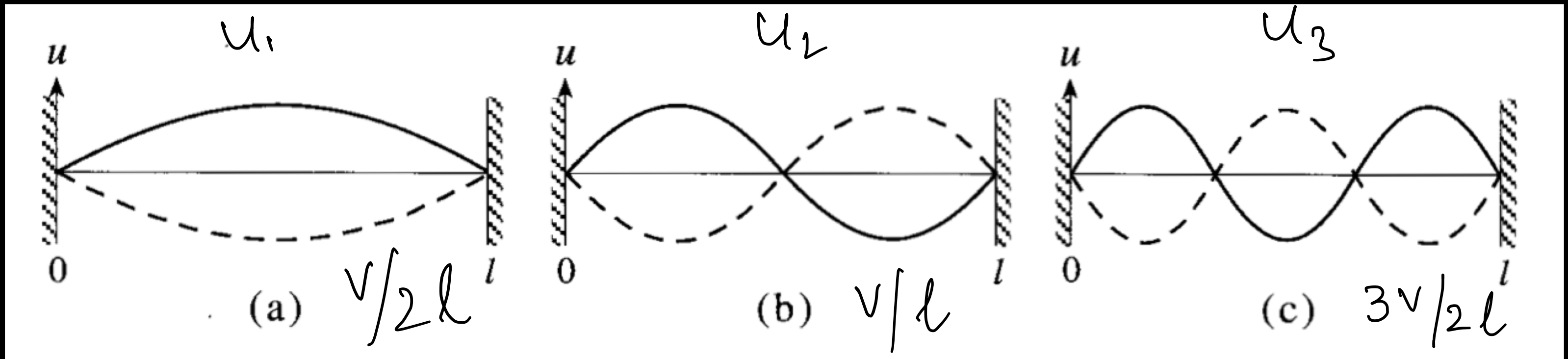


FIGURE 2.3
The first three normal modes of a vibrating string. Note that each normal mode is a standing wave and the the nth harmonic has $n-1$ nodes.

Harmonic motion frequency = vn/2l

u_1 (x,t) = fundamental mode or First harmonics, frequency = v/2l
u_2 (x,t) = 2[nd] Harmonics or 1st overtone, frequency = v/l, midpoint is called nodes. This is twice the frequency of 1[st] harmonics.
U3(x,t) = 3[rd] harmonics, frequency = 3v/2l.

So, n[th] harmonics has (n-1) nodes.

"Derivation" of Schrodinger Equation → wave eqn of quantum particle.

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Equation for finding wave function of a particle

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Solution to Schrödinger equation → wave function

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Time independent → Stationary State wavefunction

- Classical mechanics $\xrightarrow[\text{law}]{\text{fundamental}}$ $\boxed{P = mf}$
- Quantum Mechanics $\xrightarrow[\text{law}]{\text{fundamental}}$ $\boxed{\text{Schrödinger Eq}^n}$

Classical one-dimentional wave eqn →

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 u}{\partial t^2} \Rightarrow u(x,t) = \underbrace{\psi(x)}_{\text{spatial part}} \cos\omega t \quad \text{— (1)}$$

'Si'

$$\cancel{\cos(\omega t)}\frac{d^2\psi(x)}{dx^2} = -\frac{\omega^2}{v^2}\psi(x)\cancel{\cos(\omega t)}$$

part

$$\frac{d^2\psi(x)}{dx^2} + \frac{\omega^2}{v^2}\psi(x) = 0$$

$\underset{\text{ang. vel.}}{\omega} = 2\pi\underset{\text{Ang. Freq.}}{\nu}$ { $\nu\lambda = v$ }

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2}{\lambda^2}\psi(x) = 0 \quad \text{— (2)}$$

Introducing de Broglie matter wave ⇒

$$\text{Total Energy, } E = \underbrace{\frac{p^2}{2m}}_{\text{Kinetic}} + \underbrace{V(x)}_{\text{potential}}$$

$$P = \sqrt{2m(E - V(x))}$$

$\lambda = \frac{h}{p}$ de Broglie Waves, $\hbar = \frac{h}{2\pi}$

$$= \frac{h}{\sqrt{2m(E - V(x))}} \quad \text{— (3)}$$

Substituting this in eqn (2) ⇒

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi(x) = 0$$

$\left(\hbar = \frac{h}{2\pi}\right)$

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)}$$

"Time-Independent Schrödinger Eqn"

Operators

$2 \boxed{+} 3 = 5$

$2 \boxed{\times} 3 = 6$ → Operator

Do something to object

↓

Creat a new object

$\hat{O} f(x) = g(x)$

Example: $\frac{d}{dx} f(x)$ $\quad \int f(x)\,dx$ $\quad \sqrt{f(x)}$ $\quad 2f(x)$ $\quad (f(x))^n$

Linear Operator: (In Quantum Mechanics)

$\hat{O}[c_1 f_1(x) + c_2 f_2(x) + \cdots] = c_1 \hat{O} f_1(x) + c_2 \hat{O} f_2(x) + \cdots$

$\hat{O}\left(\sum_{i=1}^{n} c_i f_i(x)\right) = \sum_{i=1}^{n} c_i \hat{O} f_i(x) \quad \{c_i\} \in \mathbb{C}$

$\frac{d}{dx}(2x^2 + 3x) = 2\frac{d}{dx}(x^2) + 3\frac{d}{dx}(x)$

$\sqrt{2x^2+3x} \neq 2\sqrt{x^2} + 3\sqrt{x}$ ✗

(Classical properties) → (Liner, Quantum Mechanical Operator)

$\hat{p}$ → Momentum

$\hat{x}$ → Position

$\hat{T}$ → Kinetic Energy

$\hat{V}$ → potential Energy

$\hat{H}$ → Total Energy

"Hamiltonian"

Eigenvalues

Operator (Function) = Constant (Same function)

$\hat{A}\psi = a\psi \longrightarrow$ Eigen Equ^n

Eigen Function (ψ), Eigen Value (a)

$$\underbrace{\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\right]}_{\hat{H} \Rightarrow \text{Hamiltonian Operator}}\psi(x) = E\psi(x) \implies \boxed{\hat{H}\psi(x) = E\psi(x)}$$

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$$

E, $\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$ total Energy

T, $\hat{T} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$ Kinetic Energy

V, $\hat{V} = V(x)$ potential Energy

P_x, $\hat{P}_x = -i\hbar\frac{d}{dx}$ momentum in x direction

x, $\hat{x} = x$ x- position

$\hat{P}_x^2 = -\hbar^2\frac{d^2}{dx^2}$ momentum squared

x^2, $\hat{x}^2 = x^2$ position squared